Detecting War-Related Fires in Ukraine from Satellite Fire Data

A machine-learning approach compared with The Economist's war-fire model

Executive Summary

The proposed setup flags individual war-related fires in Ukraine, and about two in three of its alerts are war fires. The Economist's model, judged on the same labels, gets only about one in sixteen right.

The problem. Satellites recorded 694,677 fire detections in Ukraine between 24 February 2022 and 31 August 2026. Some were caused by shelling, missiles and drones. Most were field burning, wildfires and industrial heat. No dataset says which fire is which. The best-known public method, The Economist's war-fire model, flags *areas* with more fire than usual. It doesn't judge each fire on its own.

What was built. A pipeline that labels fires using public data and trains a machine-learning model (XGBoost) to score every new fire:

- A fire is labelled **war** when it is within 5 km and one day of a precisely recorded deadly conflict event (UCDP).
- A fire is labelled **normal** when it matches a peacetime pattern (recurring, agricultural, forest or industrial) *and* is far from the fighting.
- The model only sees the satellite readings and the fire's context: land type, fire history, population, season, weather. It never sees the conflict records, so it can also score fires those records missed.
- Two models are combined to make a **setup**. One knows the location and covers the area near the fighting. The other ignores location and covers the rest of Ukraine. Both are retrained every six months.

Headline results (rolling six-month retraining, four test periods, July 2024 – June 2026):

- **Precise alerts.** 65% of our setup's alerts are war fires, and it catches 66% of them. The Economist catches 79%, but only 6% of its alerts are war fires.
- **Far fewer alerts**. To catch the same number of war fires, the setup needs 7 to 13 times fewer alerts than The Economist.
- **Confidence tiers.** When both our models agree, 87% of alerts are war fires. When the setup and The Economist agree, 77% are.
- **Independent confirmation.** Far from the front, our alerts fall on the day of a nearby strike reported by VIINA (a news-based event dataset neither method was trained on) 1.7 to 2.6 times more often than chance (July 2024 – August 2026). The Economist's alerts do no better than fires picked at random.

- **Retraining is essential.** A model trained once and used more than a year later loses most of its skill as the front moves (78% of its PR-AUC for the location model). Retrained every six months, both models stay far above random in every period.
- **Context, not heat.** The fire's heat readings alone score only 1.2 times a random guess. What separates war fires is where and when they burn.

Limitations. The labels are approximations: UCDP only records events with deaths, so many strikes never become war labels. The comparison on these labels also favours our approach, because our models were trained on them. Near the front, where most fires burn, strikes are reported almost daily, so no method can be checked fire by fire and precision there is unknown. The setup has known blind spots: the northern border with Russia, strikes on factories that burned often before the war, the 30–100 km band around the front, and seasonal field burning. It also must be retrained every six months.

Recommendations. Build the production pipeline that trains both models every 6 months and use the combined setup as a fire-level detector with three confidence tiers. Keep The Economist's model alongside it as an area-level signal, especially on the northern border. Retrain with ACLED instead of UCDP as the source of war labels, which would add non-fatal strikes and cover the border strip. Before wider use, hand-check a sample of alerts and add labels for strikes on industrial sites.

1 Introduction

Fire is one of the few signs of war that can be seen from space every day. Telling war fires apart from normal fires gives near-real-time evidence of where fighting and strikes are happening.

1.1 Background

The VIIRS sensors on the S-NPP and NOAA-20 satellites pass over Ukraine several times a day. They record any unusually hot spot at about 375 m resolution, and NASA's FIRMS service publishes these detections within hours.

Since February 2022, shelling, missiles and drones have set fields, forests, buildings and fuel depots on fire. But Ukraine is also a large farming country. Stubble burning after harvest, spring grass fires, forest fires, gas flares and steel plants produce far more detections than the war does. On the satellite's readings, a shelled wheat field and a burnt stubble field look almost the same.

1.2 The existing benchmark: The Economist's war-fire model

The Economist built the best-known public method. It learns from pre-war years how much fire is normal for each place and time of year. It then flags fires in areas where activity is abnormally high. This works well as an area-level signal: it shows where unusual burning is happening.

The Economist also notes its limits. Some fires it flags are probably not war-related, many war fires are missed, and locations can be off by up to 1 km. It also doesn't say how likely each *individual* fire is to be war-related.

1.3 This project

This project builds a fire-level detector: a model that gives each fire a score for how likely it is to be war-related. Earlier studies have used fire weather indices, night-time heat sources and commercial imagery for similar tasks. Unlike them, this project classifies individual fire detections using only public data, trains on labels built from public conflict records, and checks its results against both The Economist's model and an independent dataset of strike reports.

A fire-level detector could serve journalists, researchers and organisations that monitor conflicts and their effect on civilians. They need to know *which* fires are likely war-related, and when.

Scope. Ukraine, including occupied areas and Crimea, from 24 February 2022 to 31 August 2026. Pre-war fire data from 2019 onwards is used only to describe what normal fire looks like at each place.

2 Problem Definition & Research Questions

The task is to give every satellite fire detection in Ukraine a score for how likely it is to be war-related, using only information available on the day it is detected.

Four things make this hard: no dataset records which fires the war caused, so labels must be built from conflict records and rules; only 1 in 25 labelled fires is a war fire; the front moves, so a model that learns where fighting was can fail where it is now; and war and normal fires look alike, since fighting and harvest burning both peak in summer.

Research questions

- **RQ1.** How can reliable training labels for war-related and normal fires be built from public conflict-event data and contextual rules when no ground truth exists?
- **RQ2.** Which information separates war-related from normal fires: the fire's own heat signature, or its context (place, fire history, season, weather)?
- **RQ3.** How well does a model trained on past months work on later months as the front moves, and which setup keeps it reliable?
- **RQ4.** How does the proposed approach compare with The Economist's war-fire model, against the project's labels and against independent strike reports?

3 Data Sources

The pipeline combines nine public datasets. Only the fire detections and their context reach the model. The conflict datasets are used in three other ways: to build the training labels, to measure how far each fire is from the fighting (which decides which of the two models scores it), and to check the results.

Dataset	What it provides	How it is used
NASA FIRMS, VIIRS 375 m (S-NPP and NOAA-20 satellites)	1,205,215 fire detections, January 2019 - August 2026: location, time, brightness, fire	Fires to classify, heat features, pre-war fire history

	radiative power (FRP), confidence, day or night	
UCDP Georeferenced Event Dataset (annual release plus monthly candidate releases)	Conflict events with at least one death, with date and location	War labels: Only events whose date and location UCDP records precisely (precision level 1)
The Economist war-fire model data	For each fire since 24 February 2022: local excess fire activity, population, and The Economist's war-fire flags	Model features, labelling normal fires (Section 5.3), benchmark for comparison
ESA WorldCover 2021 (10 m)	Land-cover class (cropland, forest, built-up, etc.)	Model feature, labelling normal fires (Section 5.3)
Fire Weather Index (Copernicus Emergency Management Service, daily, via Google Earth Engine)	How fire-prone the weather is each day	Model feature, labelling forest fires (Section 5.3)
VIINA territorial control	Daily control status (Ukrainian, Russian or contested) of 33,141 settlements	Distance to the front line: used for labelling and to choose which model scores each fire
VIINA event data	Conflict events geolocated from Ukrainian and Russian news reports, updated every 6 hours	Independent check of the results
Natural Earth (10 m boundaries)	Russia's land border	Distance to the Russian border
ACLED Ukraine conflict monitor	Maps and counts of conflict events	Visual cross-check only

4 Methodology Overview

The pipeline contains five sequential phases: Fire Data Collection, Feature Engineering (including the distance to the fighting), Labelling (war, normal and unknown fires), Model Training, and Scoring and Evaluation, which covers routing each fire to the right model, assigning confidence tiers and validating the results against The Economist's model and independent VIINA strike reports.

The pipeline was implemented in Python on Google Colab, using libraries including pandas, NumPy, GeoPandas, SciPy, scikit-learn, XGBoost, the Google Earth Engine API and Matplotlib. The complete implementation and replication code can be reviewed in the notebook War_Fire_Tracker_Pipeline.ipynb in the Project's GitHub Repository.

A visual schematic of this flow is shown in Figure 1. For a more detailed explanation of each phase, **see Section 5 (Detailed Pipeline Methodology)**.

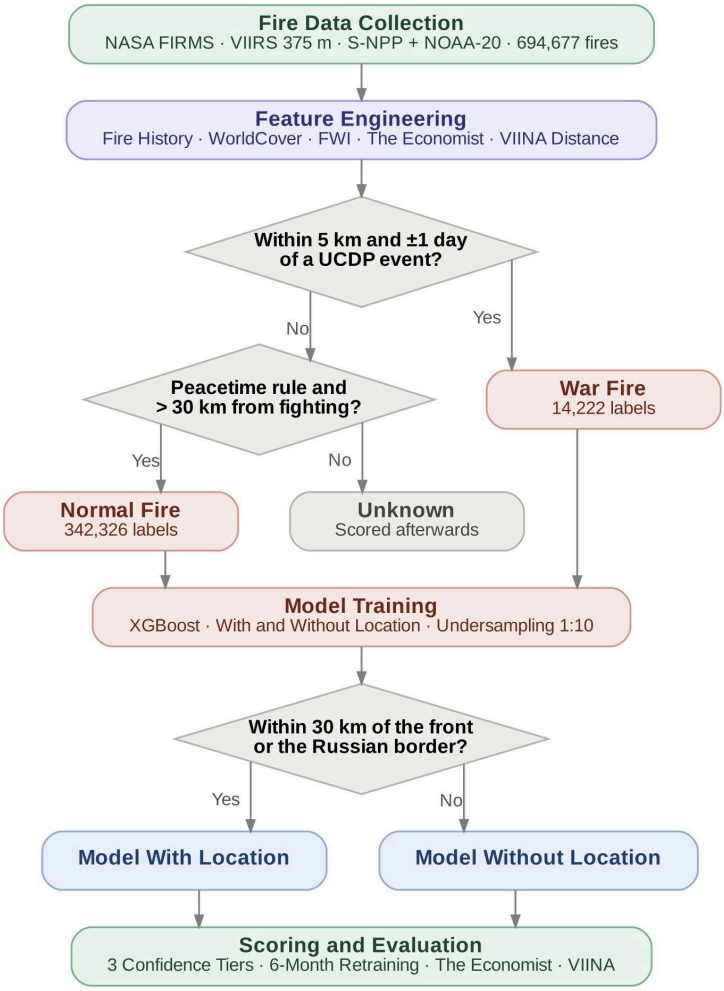


Figure 1. Pipeline Architecture Flowchart

5 Detailed Pipeline Methodology

5.1 Fire Data Preparation

- VIIRS 375 m detections from both satellites were downloaded for Ukraine from 1 January 2019 to 31 August 2026. The most recent months come from FIRMS near-real-time data. After removing exact duplicates, there are 1,205,215 detections, each with a unique `fire_id`.
- Only fires from the start of the invasion (24 February 2022) are classified: 694,678 detections. One with impossible coordinates was removed, leaving 694,677.
- The 510,537 pre-war detections are used only to describe each place's normal fire history. They are not training examples (see 5.7).

5.2 Features

The model sees fourteen features in six groups. The model "with location" uses all fourteen; the model "without location" drops latitude and longitude.

Group	Feature	Meaning	Source
Heat	brightness, bright_t31	Brightness temperature in two infrared channels (K)	FIRMS
Heat	frp	Fire radiative power: how much energy the fire gives off (MW)	FIRMS
Heat	confidence	Detection confidence	FIRMS
Place	latitude, longitude	Coordinates (for the location model only)	FIRMS
Place	land_cover_code	Land-cover class at the fire's position	ESA WorldCover
Place	pop_exact	Population at the fire's location	The Economist
Fire history	pre_war_annual_freq	Average detections per year in the same ~1.1 km grid cell before the invasion	FIRMS, 2019 – Feb 2022
The Economist's signal	excess_fire, fire_in_window	How much more fire there is in the area than expected for that place and time of year	The Economist
Time	month, daynight	Season; day or night satellite pass	FIRMS
Weather	fwi	Fire Weather Index on the day	Copernicus

Merging The Economist's data. Each fire is matched to the nearest Economist detection within 200 m on the same day; 85.5% of post-war fires find a match. Unmatched fires keep missing values (NaN), which XGBoost handles on its own. In the training data, The Economist's features are missing for 38,059 fires and FWI is missing for 553.

Distances that are not features. Three more columns describe where a fire is relative to the fighting. They are used to build labels and to decide which model to ask, but the models never see them.

1. **Distance to the contact line.** For every day, VIINA gives the control status of each settlement. For each fire, the pipeline measures the distance to the nearest Ukrainian-held (d_UA), Russian-held (d_RU) and contested (d_contested) settlement on that day:

$$d_{\text{front}} = \min\left(d_{\text{contested}},\ \max(d_{\text{UA}},\ d_{\text{RU}})\right)$$

The larger of d_UA and d_RU is the distance across to the other side; contested settlements count as front line from both sides. Gaps of up to 7 days in VIINA's data are filled with the last known status. The median fire is 47.9 km from the contact line.

2. **Side of the front.** A fire is on the Russian-held side when d_RU is smaller than d_UA. 41.1% of fires are on the Russian-held side.

3. **Distance to the Russian border**, from Natural Earth boundaries. It covers the northern regions (Sumy, Kharkiv, Chernihiv), where there is no contact line but Russia shells across the border.

5.3 Labelling

Each fire goes through the checks below in order; the first one that fits decides its label.

1. **War fire.** The fire is within 5 km and one day of a UCDP event whose date and location UCDP records precisely (precision level 1). This matches 16,133 fires. **Industrial correction:** 2,024 of those war fire matches sit at permanent industrial sites, where routine heat can fall near an event by coincidence. If the fire's FRP is no higher than that site's pre-war 95th percentile, it is moved to unknown (1,911 fires). If it is hotter than usual, it stays a war fire (113 fires). This leaves **14,222** war fires.
2. **Normal fire.** The fire matches one of four peacetime rules *and* is far from the fighting (table below). This matches **342,326** fires.
3. **Unknown.** Everything else: **338,129** fires. They are left out of training and scored afterwards.

The training set therefore has 356,548 labelled fires: one war fire for every 24 normal fires. War fires were rarest in 2026 (1.7% of labelled fires, against 2.5% to 6.7% in earlier years). Most unknown fires are near the fighting, where no normal label is allowed: 265,654 near the contact line and 52,055 in the northern border strip. The war labels are also only a confirmed subset of war fires: UCDP records only events with at least one death, and only 2.4% of The Economist's war fires fall within 5 km and a day of a precise UCDP event, and only 15.1% of those events have an Economist war fire nearby.

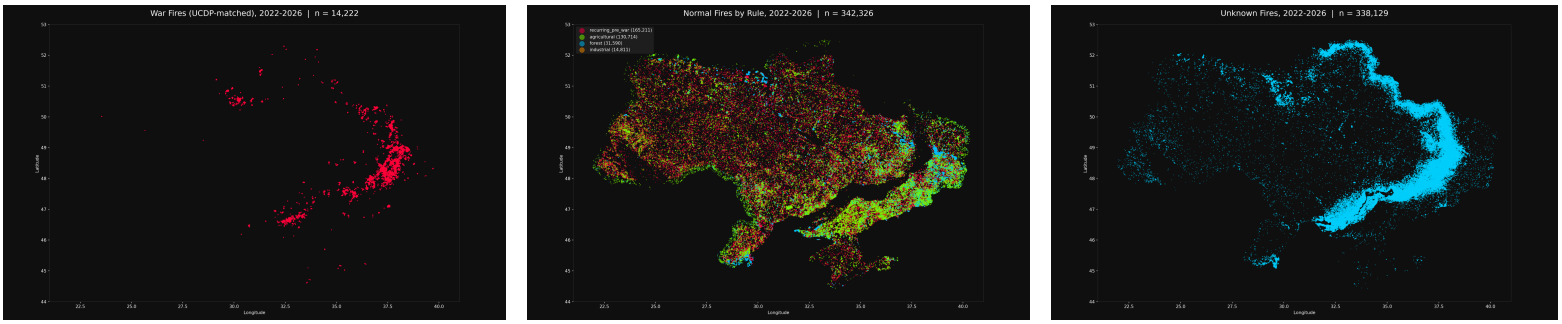


Figure 2. War, normal (coloured by rule) and unknown fires, 24 February 2022 – 31 August 2026.

"Far from the fighting" means more than 30 km from the contact line, and, on the Ukrainian-held side, also more than 30 km from the Russian border. Mortars, tube artillery, Grad rockets and FPV drones mostly reach 20–30 km. UCDP misses many war fires near the fighting, so a "normal" label there could be a mislabelled war fire.

Normal rule	Condition (always plus far from the fighting)	Fires
Recurring	excess_fire ≤ 0: no more fire in the area than expected for that place and time of year	165,211
Agricultural	Cropland, grassland or shrubland, in any month	130,714
Forest	Tree cover, herbaceous wetland or bare/sparse vegetation, with FWI above 15	31,590

Industrial	More than 50 detections a year before the war in the same ~1.1 km cell, on built-up land	14,811

5.4 Model and Training

- **Model.** XGBoost (gradient-boosted decision trees): up to 1,000 trees, learning rate 0.05, depth 6. Training stops early when validation PR-AUC hasn't improved for 50 rounds.
- **Class imbalance.** Normal fires in the training data are randomly sampled down to 10 for every war fire. Validation and test data keep their real ratio. In earlier trials this worked better than weighting the classes.
- **Alert threshold.** The score above which a fire becomes an alert is chosen on validation data (best F1), then frozen before testing. Scores rank fires; they are not probabilities, because undersampling shifts them.
- **Feature-set comparison.** Six versions are trained to see what each group of features adds: all features, heat only, context only, all without The Economist's features, all without location, and all without location or The Economist's features.

5.5 Evaluation Design

Three tests are used. The rolling test is the main one, because it copies how the setup would be used.

Test	Trained on	Validation on	Tested on	Question it answers
Chronological	24 Feb 2022 – Dec 2024 (214,916 fires)	2025 (75,688)	Jan–Aug 2026 (65,944)	Can a model trained on the past score fires 12–20 months later?
Spatial	70% of 0.5° blocks, about 55 × 37 km each (249,207)	15% of blocks (50,496)	15% of blocks (56,845)	Does it work in areas it hasn't seen before?
Rolling 6-month retraining	Everything before the validation window	The 3 months before each cutoff	The 6 months after each cutoff	How would it perform if retrained every six months?

The rolling cutoffs are 1 July 2024, 1 January 2025, 1 July 2025 and 1 January 2026, giving four test periods from July 2024 to June 2026.

How performance is measured.

- **Precision**: of the fires flagged as war, the share that are war fires.
- **Recall** ("caught"): of all war fires, the share that were flagged.
- **F1**: one number combining precision and recall; it is high only when both are.

- **PR-AUC**: how well the model ranks war fires above normal fires, over every possible threshold. A random guess scores the share of war fires in the test data (between 0.025 and 0.064 in the four rolling test periods), so each PR-AUC is read against that baseline.
- **Alerts**: how many fires a method flags, and how many alerts it needs per war fire caught.

5.6 The proposed setup

Two models are retrained every six months and combined by distance:

- **Model with location** (all 14 features) scores fires in the near-fighting zone: within 30 km of the contact line, or on Ukrainian-held land within 30 km of the Russian border.
- **Model without location** (12 features) scores every other fire.
- The two models have different thresholds, so each fire's score is divided by its model's threshold. A result of 1 or more is an alert.

Alerts come in three confidence tiers:

Tier	A fire is in this tier when…	Use
1 · High confidence	Both models flag it	Fewest but most reliable alerts
2 · Strong	The combined setup and The Economist's war_fire both flag it	Works only while The Economist publishes its data
3 · Standard	The combined setup flags it	The main balanced detector

5.7 Key design decisions

The pipeline went through several rounds of review. The table lists the problems found along the way and how each was fixed; all results in this report use the final version.

Problem found	Why it mattered	Change made
Era leakage	The Economist's data starts on 24 Feb 2022. Filling pre-war gaps with fixed values (e.g. population 0) let the model recognise the era instead of the fire.	Pre-war fires dropped from training; pre-war data used only for baselines
Wrong normal labels near the front	UCDP misses many war fires near the fighting, so fires labelled normal there were often war fires.	"Far from the fighting" condition added to all four normal rules
Contact line measured to the wrong side	Distance to the nearest Russian-held settlement is about 0 for every fire in occupied areas, including Crimea.	Distance to where the two sides meet, with contested settlements counted from both sides; Russian border added
Location shortcut	With no normal labels near the	A second model without

	front, a model with coordinates learned "near the front = war" and failed once the front moved.	coordinates; models combined by distance and retrained every 6 months
Seasonal shortcut	The agricultural rule first applied only in harvest months (August–November), so the model learned "spring field fire = war".	Rule 2 applied in every month instead
Factory heat labelled war	Routine heat at steel plants and gas flares near a UCDP event was labelled war.	FRP check against each site's pre-war range: 1,911 labels moved to unknown, 113 kept
Leaky evaluation	Rounded coordinates mismatched The Economist's data; small 0.1° test blocks let neighbouring areas leak; thresholds were once picked by looking at test results.	200 m same-day matching; 0.5° blocks; thresholds chosen on validation only and frozen

The earlier versions and experiments behind these decisions, with their original outputs, are kept in the [Project's GitHub Repository] in a separate notebook, War_Fire_Tracker_Experiments_Archive.ipynb.

6 Results

6.1 Which information matters

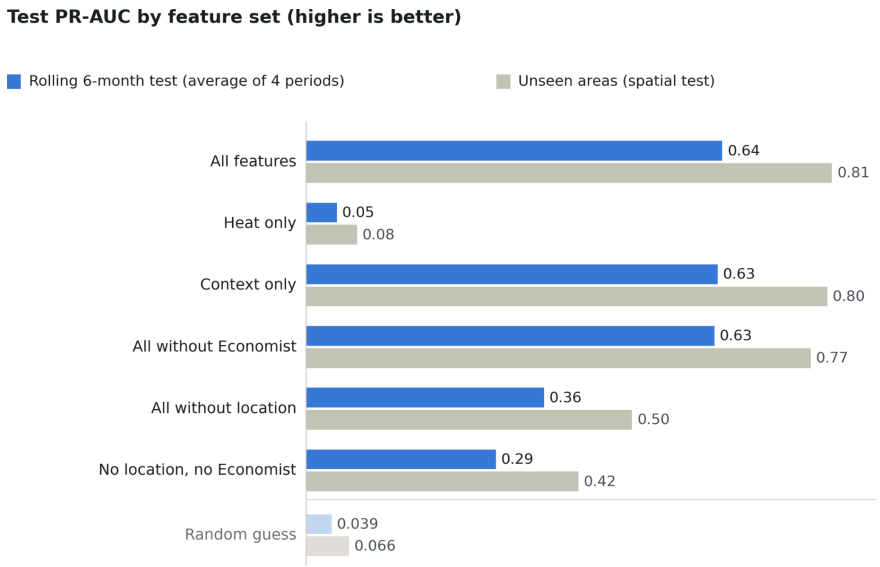


Figure 3. Test PR-AUC for six feature sets: rolling 6-month test (average of the four periods) and spatial test.

The fire's heat readings alone are barely better than a random guess, 1.2 times random in both tests, and adding them to the context barely changes the score (0.638 against 0.630 in the rolling test). The signal comes from the fire's context. With six-month retraining, location helps: removing the coordinates

lowers PR-AUC from 0.64 to 0.36, because a recently retrained model knows where the front is now. Section 6.2 shows that this advantage disappears when the model is not retrained. The Economist's features add little when coordinates are present (0.638 to 0.625) but matter more without them (0.365 to 0.291), because they are then the model's only up-to-date signal of where fighting is happening. In the spatial test, the location model's 0.81 is flattering: the test blocks sit next to training blocks on the same dates, and an earlier block draw gave the same model only 0.57.

What the two models rely on (share of total gain in the rolling test):

Feature group	Model without location	Model with location
Location (latitude, longitude)	-	44.5%
Type of place (population, land cover)	37.1%	21.3%
Time (month, day or night)	19.1%	14.0%
Fire history (pre-war fire frequency)	18.9%	7.1%
The Economist's signal (excess fire, fire in window)	13.1%	6.9%
Heat (brightness, FRP, confidence)	7.7%	3.8%
Weather (Fire Weather Index)	4.2%	2.0%

The model with location relies on coordinates for almost half of its gain (about 45%). Without them, the type of place a fire burns in, its timing and its fire history carry three quarters of the gain, and The Economist's signal roughly doubles in share (7% to 13%).

6.2 Performance over time

A model trained once loses most of its skill as the front moves. Retrained every six months, both models stay far above random in every period.

Trained once, tested much later. In the chronological test, the models were trained on data up to December 2024 and tested on January–August 2026, 12 to 20 months later.

Model	PR-AUC, validation (2025)	PR-AUC, test (2026)	Change
With location	0.604	0.133	-78%
Without location	0.442	0.277	-37%

Both models score much lower on 2026 than on 2025, but the location model loses far more: 78% of its PR-AUC, against 37% for the model without location. By 2026 the front had moved away from where the location model learned it, so the model without location scored about twice as high (0.277 against 0.133).

Part of the drop affects both models: war fires were rarer in 2026 (1.7% of labelled fires, against 4.8% in 2025), which lowers PR-AUC even when a model is just as good.

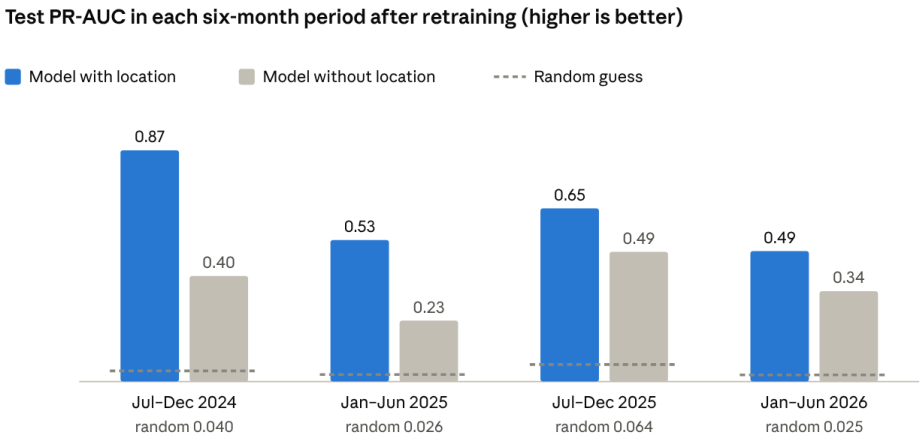


Figure 4. Rolling six-month retraining, July 2024 – June 2026, test PR-AUC per period

With regular retraining, the picture reverses: the location model ranks better in every period, at 1.3 to 2.3 times the PR-AUC of the model without location, because six months is too short for the front to move far. Both models are weaker in January to June, when there are fewer war fires, more spring field burning, and thresholds chosen on autumn data. The location model still misses the far deep strikes, which is why the proposed setup combines both models (Section 6.3).

6.3 Choosing the setup

Overall, combining the two models by distance performs about the same as the location model alone. Its advantage is beyond 30 km from the front, where it catches more than twice as many war fires. The table below covers the four rolling test periods (6,744 war fires, July 2024 – June 2026).

Setup	War fires caught	Precision	False Alarms
Both models flag	29%	87%	281
Model with location only	65%	67%	2,145
Model without location only	32%	49%	2,287
Combined by distance (proposed setup)	67%	66%	2,287
Either model flags	69%	53%	4,151

The combined setup has the same false alarms as the model without location, because false alarms can only be counted more than 30 km from the front, where the setup uses that model.

The two models are good at different things:

Distance to the contact line	War fires	Caught: with location	Caught: without location
0–30 km	6,260	69%	33%
30–100 km	324	13%	26%
100–200 km	70	30%	17%
More than 200 km	90	6%	60%

The location model works as a front-line detector: it knows where fighting has recently been, but it misses almost every deep strike far from the front (5 of 90). The model without location is the opposite, a deep-strike detector that catches 54 of the 90 war fires more than 200 km from the front. Combining them keeps both strengths. Beyond 30 km, the combined setup catches 163 war fires against 68 for the location model alone, with no loss of precision. Flagging a fire when either model flags it isn't worth it: it catches only 2 percentage points more war fires but nearly doubles the false alarms, about 13 extra false alarms for each extra war fire. Flagging only when both models agree is the natural top tier. It was the most precise setup in every period (99%, 81%, 89% and 62%), and since all of its alerts are also combined alerts, it marks the most certain ones rather than forming a separate system.

6.4 Comparison with The Economist

The two methods answer different questions. The Economist's model asks "where is fire activity abnormal?" and gives an area-level signal. The proposed setup asks "is this particular fire likely war-related?" Three comparisons were made. The independent check with VIINA is the fairest, because neither method learned from it. The comparisons on labelled and unknown fires support it.

6.4.1 Independent check with VIINA strike reports

Far from the front, the setup's alerts burn on the day of a nearby reported strike 1.7 times more often than chance. With tighter matching it rises to 2.6 times. The Economist's alerts show no such link.

How the check works. VIINA's news-based events tagged as artillery, air strikes or drones (52,421 in 2024–2026) are compared with 440,377 fires from July 2024 to August 2026, labelled and unknown.

- A fire matches when such an event lies within 5 km of it, on the same day, the day before, or the day after. Tighter (3 km) and looser (10 km) settings are also tested, because VIINA places events at settlement centres, not at the exact impact point.
- Near the front, strikes are reported almost every day, so almost any fire will have one nearby by chance. To measure that chance level, each fire is also matched against events from 30 days earlier and later (a placebo).
- **Lift** = match rate on the real date ÷ match rate on the placebo dates. A lift near 1 means the flagged fires sit in often-struck places but not on strike days; above 1 means they burn on strike days more often than chance.

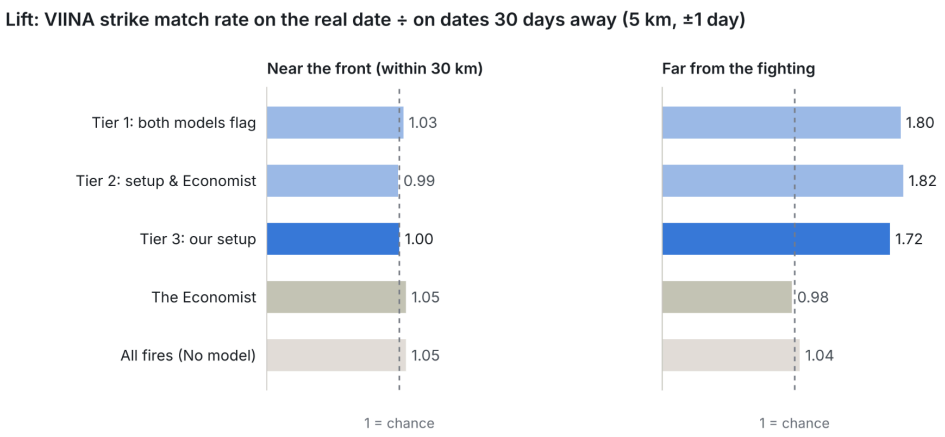


Figure 5. VIINA check by zone, July 2024 – August 2026, 440,377 fires, 52,421 VIINA strike events

Far from the fighting, our alerts line up with strike days: all three tiers score 1.7 to 1.8, while The Economist (0.98) does no better than fires picked at random (1.04). Even at the same number of alerts as The Economist, the setup still scores 1.18 against 0.98. Near the front, VIINA can't separate the methods, because strikes are reported near most villages almost every day, so every method, and every fire, scores about 1. The northern border strip can't be judged either: only about 1% of fires there match any VIINA event.

The far-from-fighting result holds at every setting, and grows as matching gets tighter, as expected if the alerts are tied to specific strikes:

Matching rule	Tier 1: both models flag	Tier 2: setup & Economist	Tier 3: our setup	The Economist	All fires (No model)
3 km, same day	1.66	3.42	2.57	1.21	1.16
3 km, ±1 day	2.04	3.23	2.42	1.06	1.07
5 km, ±1 day	1.80	1.82	1.72	0.98	1.04
10 km, ±1 day	1.28	1.27	1.26	0.97	1.02

Our alerts also sit in places where VIINA reports more strikes, but part of that comes from learning UCDP's conflict areas, which is why the timing check above is the main test.

Are the "false alarms" really false? A false alarm here is a flagged fire that our labels call normal. Matching them against VIINA (3 km, ±1 day):

Group	Fires	VIINA strike nearby	Placebo	Lift	Likely real war fires
Our false alarms	2,598	4.0%	2.4%	1.65	about 41

Of these, on forest land	703	4.7%	1.0%	4.7	about 26
The Economist's false alarms	105,057	0.6%	0.6%	0.97	about 0
Normal fires nobody flagged	83,696	0.9%	0.9%	1.01	-

About 41 of our 2,598 false alarms (1.6%) look like real strikes, mostly forest fires that may have been started by strikes or falling drone and missile debris. Our measured precision is therefore slightly too low, but only slightly. The Economist's false alarms really are false: they have fewer strikes nearby than normal fires nobody flagged. It raises about 40 times as many false alarms as the setup on the same fires.

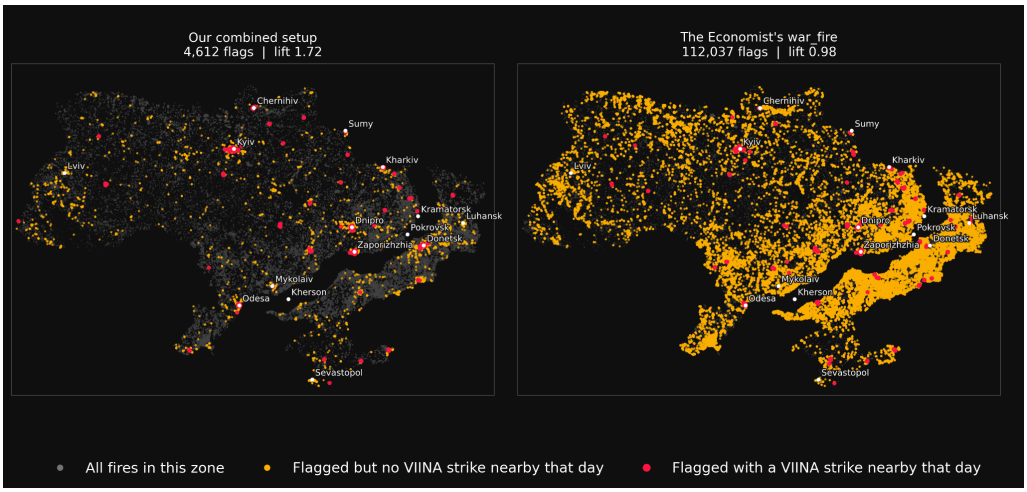


Figure 6. VIINA check far from the fighting: flagged fires with and without a VIINA strike nearby.

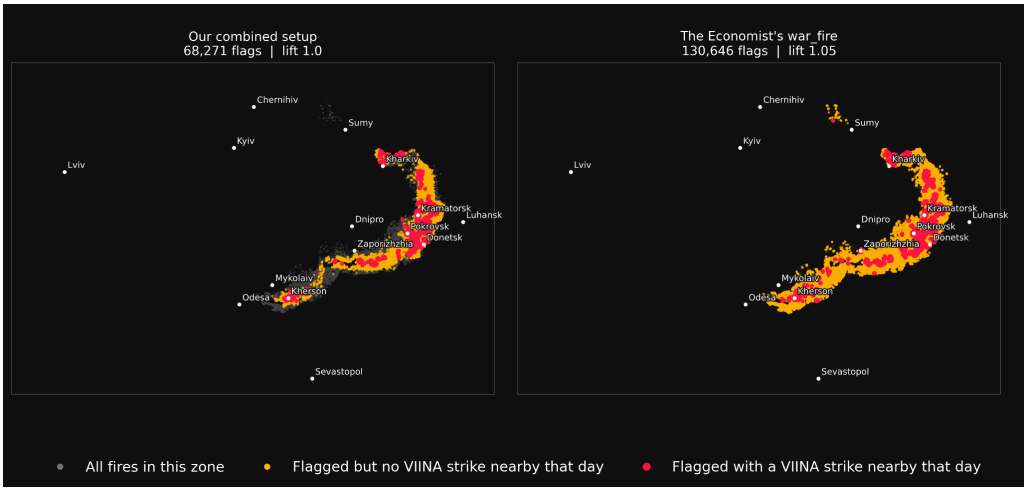


Figure 7. VIINA check near the front (within 30 km), where strikes are reported almost daily and no method stands out.

6.4.2 Against the project's labels

On the labelled test fires, the combined setup's alerts are about 11 times more precise than The Economist's. The Economist catches more war fires, but only by flagging most fires. This comparison favours our setup by design, because the labels come from UCDP and our models were trained on them. The comparison uses the 151,764 labelled fires from the four rolling test periods that also appear in The Economist's data (4.3% of them are war fires).

Method	Precision	War fires caught	Alerts
Tier 1: both models flag	87.1%	29.3%	2,176
Tier 2: combined setup and The Economist agree	77%	53%	4,421
Tier 3: combined setup	65.4%	66.2%	6,546
The Economist war_fire	6.1%	79.0%	83,644
The Economist war_fire_restrictive (tighter version)	5.3%	48.3%	58,805

At the same detection rate. The Economist's model gives a fixed yes or no, while our threshold can be moved. Lowering our threshold until the setup catches as many war fires as The Economist gives a like-for-like comparison:

To catch as many war fires as…	Our alerts	The Economist's alerts	Our setup needs
war_fire (79% caught)	11,990	83,644	7× fewer alerts
war_fire_restrictive (48% caught)	4,630	58,805	13× fewer alerts

The setup is more precise than The Economist in every one of the four periods (35–91% against 3–9%). The two methods also make different false alarms: The Economist's are mostly farmland (41,352) and forest (15,478), plus 6,704 at industrial sites, while ours are mostly recurring fires around cities and occupied areas (1,148), with 664 forest, 452 farmland and none industrial. Used together, they complement each other and catch 93% of war fires. The Economist finds 1,704 war fires that our setup misses, but only by raising 77,519 false alarms that ours doesn't.

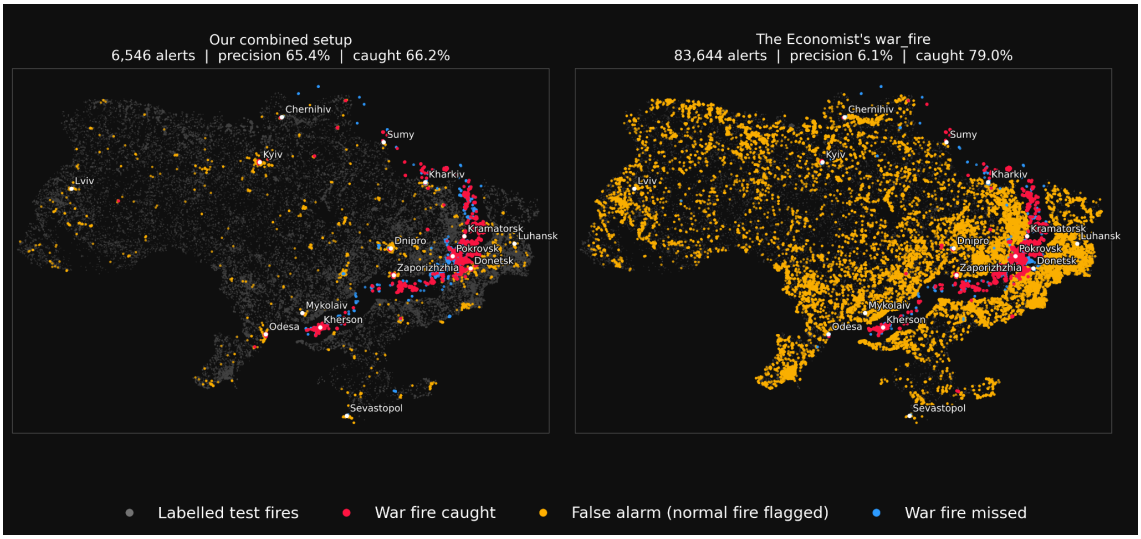


Figure 8. Labelled test fires in the four rolling test periods (July 2024 to June 2026). Red: war fires caught; orange: false alarms; blue: war fires missed.

6.4.3 On the unknown fires

Among fires the rules couldn't label, the combined setup flags about a third and The Economist about three-quarters. Neither share can be checked directly. The clearest difference is on routine factory heat.

The comparison covers 195,294 unknown fires from July 2024 to August 2026 that also appear in The Economist's data. Like the other results, it uses the rolling six-month retraining: each six-month period is scored by models trained only on data from before it. July–August 2026 is scored by one extra model retrained on 1 July 2026.

Method	All unknown fires	Near the fighting	Far from the fighting	Factory heat moved out of the war labels
Tier 1: both models flag	4.7%	4.8%	2.1%	0%
Tier 2: combined setup and The Economist agree	25.6%	26.1%	15.2%	0.4%
Tier 3: combined setup	34.2%	34.8%	21.0%	2.1%
The Economist war_fire	77.3%	77.3%	76.3%	56.4%

On routine factory heat, the setup flags only 2% of the 1,118 factory fires that the industrial correction moved from war to unknown, while The Economist flags 56%. Far from the fighting, 56% of the setup's flags are on built-up land, which fits strikes on towns, whereas The Economist also flags many wetland fires, such as reed burning in the Danube delta. The setup's flags also follow the fighting as it moves: in

each period, its flagged areas sit on ACLED Ukraine Conflict Monitor's densest battle zones and shift with them, from a cluster around Sumy in early 2025 to a block on the Zaporizhzhia–Dnipropetrovsk front in mid-2026, which suggests the models aren't just remembering old front lines.

The northern border, however, is a blind spot. The Economist flags 86% of unknown fires more than 200 km from the contact line and the setup only 4.3%, and many of these fires lie along the Sumy and Chernihiv borders, where ACLED Ukraine Conflict Monitor shows heavy shelling. The cause is the labels: the border strip has 0.5 war labels per 100 unknown fires, against 4.9 near the contact line, and sending border fires to the location model barely changed this. Finally, from June to August 2026 the setup's flags rose 5.6 times and The Economist's 4.4 times, while ACLED's monthly event count rose only 16%. Shelling does ignite dry fields more easily in August, but a rise this large is mostly harvest burning, picked up by the location model.

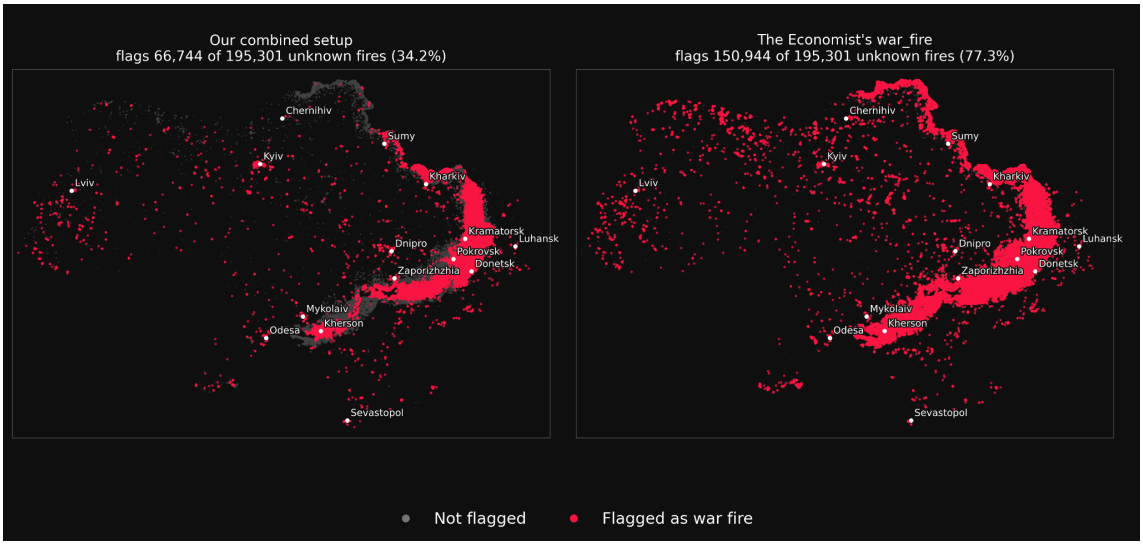


Figure 9. Unknown fires, July 2024 to August 2026, scored with six-month retraining. Red: fires flagged as war-related by each method.

7 Limitations

The labels. The labels only approximate the truth: "war" means a fire was near a UCDP deadly event, and "normal" means it matched one of the rules, so every metric measures agreement with these labels rather than with what really happened. UCDP also sees only part of the war: it records only events with deaths, and at best it locates them to a named town or village, represented by a single point; streets and parts of towns are never coded. War labels therefore cluster near populated places, so the model's heavy use of population partly reflects where UCDP places its events, not only where war fires really happen. Moreover, the rules that define normal fires use some of the same information the model sees (fire history, land cover and The Economist's excess fire). The model can therefore partly score well by learning these rules back, rather than by recognising war fires, which makes the test scores somewhat optimistic. The industrial sites show this clearly: the model raised no false alarms there, but caught only about 10% of the real strikes at those sites. Finally, the 2026 UCDP data is provisional, so events missing from it make some correct alerts count as false, and the comparison on the labelled data favours our setup, because our models were trained on these labels while The Economist's wasn't.

What can't be measured. Precision near the front is unknown. No fire within 30 km of the front is labelled normal, so false alarms there can't be counted. The independent VIINA check can't help either, because strikes are reported near the front almost every day and every method scores a lift of about 1.

Blind spots. The models are weakest in four places:

1. The northern border strip, which has only 0.5 war labels per 100 unknown fires, against 4.9 near the contact line.
2. Strikes on sites that burned often before the war.
3. The 30–100 km band, which the models treat as mostly normal because normal labels start at 30 km (only 13% and 26% of its war fires were caught).
4. Seasonal burning. Harvest fires fool the location model in summer, spring burning in the west fools the model without location, and January–June periods are clearly weaker.

Practical limits. Scores rank fires but aren't probabilities, and in the simulation the thresholds were chosen on three-month validation windows from a different season. The setup relies on The Economist publishing its data: without those features, the model without location drops from 0.365 to 0.291 PR-AUC in the rolling test. Satellites also miss fires that are short, small or under cloud, and detection positions are approximate (375 m pixels). The setup must be retrained every six months.

8 Recommendations and Conclusion

8.1 Conclusion (Answering the 4 RQs from section 2)

- **RQ1.** Usable labels can be built from public data. The price is two gaps: no normal labels near the fighting, and very few war labels on the northern border.
- **RQ2.** Context is what matters, not heat. Heat readings alone score 0.05 PR-AUC against 0.04 for a random guess. The type of place, its fire history, the time of day and The Economist's activity signal carry the information.
- **RQ3.** A model trained once and used for a year or more loses most of its skill as the front moves. Retrained every six months, the models stay far above random in every period. Combining them by distance keeps the location model's strength at the front and adds the deep strikes.
- **RQ4.** To catch the same number of war fires, the setup needs 7 to 13 times fewer alerts than The Economist. Far from the front, its alerts line up with independently reported strikes 1.7 to 2.6 times more often than chance; The Economist's don't. Near the front, neither method can be checked, and The Economist catches more war fires and covers the northern border.

8.2 Recommendations

1. **Adopt the combined setup as the main fire-level detector, with three confidence tiers:**
    - Tier 1 (both models flag): 87% precision, 29% of war fires caught.
    - Tier 2 (combined setup and The Economist agree): 77% precision, 53% caught.
    - Tier 3 (combined setup): 65% precision, 66% caught.
2. **Keep The Economist's model alongside it** as an area-level signal, especially on the northern border, where our models are blind.

3. **Build the production pipeline.** Train both models on all labelled data, choose thresholds on a 12-month validation window, retrain every six months, and check each new model on the months since the last one before switching.
4. **Hand-check a sample of alerts,** especially near the front, using news reports or high-resolution imagery. This is the only way to measure precision where the labels can't.
5. **Retrain with ACLED instead of UCDP as the source of war labels.** UCDP only records events where people died, so most shelling and drone strikes never become war labels. ACLED records these events whether or not anyone was killed, and tags each one by type (shelling, artillery, air or drone strike), so the labels could use only the events likely to start fires. This would also help in the northern border strip, where ACLED shows heavy shelling but UCDP gives almost no labels. ACLED's research access is needed, and the new labels should first be tested with the same placebo check used for VIINA, to make sure the matches are not just chance in places that are struck every day.
6. **Add war labels where they are still missing**, especially for strikes on industrial sites, for example with a feature comparing each fire's heat with its site's normal range. VIINA alone won't fill the gap: used as a label source, most of its matches would be chance, adding only about 300 real war labels.
7. **Further experiments:** keep pre-war fires in training with properly computed features, and test the model without location in another conflict. It doesn't depend on Ukraine's coordinates, so it is the version most likely to transfer.

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